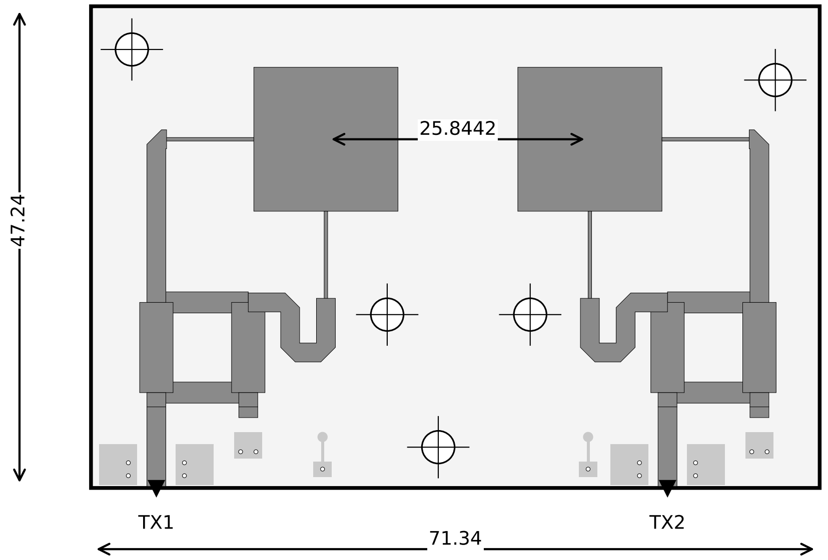
5.8 GHz 2x2 MIMO FMCW RADAR ARRAY - ZYF300CA-P PTFE

OUTLINE AND DRILL PLAN 71.34198 x 47.237 mm
(dimensions in mm, origin at lower-left)



FABRICATION NOTES

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1.  MATERIAL: ZYF300CA-P PTFE, dielectric thickness 0.76 mm +/-10%.
    Design permittivity er = 3.0 (microstrip design value), tan d = 0.0018.
    No substitution without approval: the patch dimensions are tuned to this permittivity.
2.  LAYERS: 2 copper layers. Layer 1 = antenna and feed network. Layer 2 = SOLID ground.
    The ground plane must be continuous. Do NOT add thieving, hatching or venting to layer 2.
3.  COPPER: 1 oz (35 um) finished on both layers.
4.  FINISH: IMMERSION SILVER per IPC-4553.  DO NOT SUBSTITUTE ENIG.
    At 5.8 GHz the current reaches only 0.87 um into the metal, so
    ENIG's 3 um of nickel would carry all of it, at 2.7 times the
    resistance of copper -- 4.3 per cent of detection range.
    Silver tarnish is a semiconductor and carries no RF current:
    a blackened board measures the same as a fresh one.
5.  SOLDER MASK: layer 1 has deliberate openings over the radiating elements and the
    four end-launch transitions - mask there would detune the patches and the launches.
    Mask on layer 2 as normal. Green LPI.
6.  SILKSCREEN: white, layer 1 and layer 2 as supplied. Do not print silkscreen on bare copper.
7.  IMPEDANCE: single-ended 50 ohm +/-10% on the 1.854 mm traces, referenced to layer 2.
    Impedance control is requested; adjust trace width if your stack requires it, and report
    the width used. Do NOT alter the patch or transformer geometry.
8.  ETCH TOLERANCE: +/-0.02 mm on conductor width. Minimum feature on this board is 0.341 mm
    (the quarter-wave transformers) - confirm you can hold it before starting.
9.  HOLES: 14 x 0.40 mm plated stitching vias; 5 x 3.20 mm plated mounting holes.
10. OUTLINE: routed, +/-0.15 mm. Copper reaches the board edge at the four launch sites by design.
11. NO panelisation tabs across the antenna areas. V-score not permitted.
12. ELECTRICAL TEST: 100% netlist test.
13. The mask openings in note 5 expose adjacent copper of different nets.
    That is intended on an antenna board and is the only rule waived in
    the DRC report; everything else passes at error AND warning severity.
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STACKUP

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  +-------------------------------------------+  layer 1  35 um Cu, immersion silver, mask open over antennas
  |  ZYF300CA-P PTFE      0.760 mm      |  er 3.0, tan d 0.0018
  +-------------------------------------------+  layer 2  35 um Cu, SOLID ground, masked
     finished thickness approx 0.83 mm
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DRILL TABLE

SIZE	QTY	PLATED	NOTE
0.40 mm	14	yes	ground stitching
3.20 mm	5	yes	M3 mounting, bonded to ground

ELEMENT POSITIONS (board origin at lower-left, mm)

these are the phase centres to enter into the beamforming code

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TX1  RHCP  x  22.9990   y  34.1950
TX2  RHCP  x  48.8432   y  34.1950   (mirrored)
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CRITICAL RF DIMENSIONS (do not modify)

patch	14.1135 x 14.1135 mm
element pitch	25.8442 mm (half wave in air at 5.800 GHz)
50 ohm trace width	1.8535 mm
transformer	0.3414 mm wide x 8.5422 mm long (110 ohm)
hybrid arms	8.8450 mm (vertical) / 8.9982 mm (horizontal)
launch ground gap	0.9699 mm each side